

1 SOURCE RELIABILITY



calibration $K_s \cdot K_t \cdot T_s \rightarrow t$

DEPTH RELIABILITY ENCODER

full-res residual $\rightarrow$ $\frac{1}{4}$ -res context



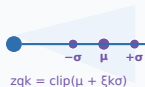
μ mean $\cdot \sigma$ scale $\cdot n$ normal
 c confidence $\cdot F_s$ feature



2 DISTRIBUTIONAL SURFACE SPLATTING

$$A_s = [I_s, F_s]$$

DETERMINISTIC σ SAMPLES

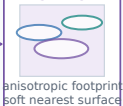


CALIBRATED PROJECTION

$K_s \rightarrow T_s \rightarrow v \rightarrow K_t$

Jacobian + tilt

TARGET-PLANE SPLATS



ONE RASTERIZATION RETURNS BOTH warped evidence + visibility statistics



target reliability $\kappa = f(C, V, H)$

3 VISIBILITY-GUIDED ROUTING



renderer-derived condition h

warped evidence

BOUNDED ROUTER

physics prior g^0 + learned Δg

$g \in [0, 1]$

TRANSPORT EXPERT

warped evidence + bounded ΔRGB

SYNTHESIS EXPERT

local conv + FFT
 $\text{RGB} \cdot \text{depth} \cdot \text{risk}$

SOFT FUSION

$(1-g)$ transport + g synthesis

4 FUSED OUTPUT

FUSED OUTPUT

blended RGB-D + predicted risk



TRAINING ONLY · DASHED

target RGB-D + mask
supervision; not an inference input

$L_{\text{rgb}} \cdot L_{\text{depth}} \cdot L_{\text{router}} \cdot L_{\text{risk}}$

BACKWARD DSS
 $T_t \rightarrow s$

source-view cycle L_{cyc}

predicted RGB-D + risk